Ori Mandarin MVP Research Report

Executive Summary

Ori already has the bones of a serious language-learning product. The public landing page promises a focused Mandarin app with **117 lessons**, **278 words**, and **8 exercise types**, and it offers a **free demo lesson** before sign-in. The public path page reinforces that promise with a structured course map showing **21 seals** and **117 steps**, split between a **Starter** track and an **HSK 1** track. The demo lesson is not just marketing copy: the public lesson route exposes a real **17-step** lesson with a listening-first exercise and browser-based Mandarin voice playback. In other words, Ori is not a vague prototype; it is already an exercise engine with public proof of curriculum breadth and lesson flow. [1]

The biggest growth limiter is not curriculum quantity. It is the gap between **promised momentum** and **felt momentum**. Today, the public experience contains several “thin” states: the `/auth` route resolves to a loading shell in the crawler snapshot, the `/review` route exposes only “Loading review…,” the profile route is extremely sparse, and the vocabulary page shows a visible pinyin rendering defect where tone-marked syllables are spaced out character-by-character, for example `y ī`, `s h à n g`, and `Z h ō n g g u ó`. Those issues weaken trust precisely where a learning app needs to feel polished and dependable. [2]

The strongest recommendation is to treat the next phase as **MVP+ habit formation**, not as “more course content.” Duolingo’s own public materials emphasize that habit-forming mechanics such as streaks, XP, quests, friend systems, and a guided path are meant to keep people returning, while its newer product updates show a shift away from pure XP grinding toward **quest-based**, **social**, and **learning-aligned** progress loops. Memrise’s recent public writing points in a similar direction, but with an important nuance: it now emphasizes **consistency dashboards** and progress visibility “without pressure or guilt.” Anki remains the clearest public reference for review rigor: active recall, spaced repetition, workload smoothing, “most lapsed” prioritization, and leech detection. Habitica is the strongest reference for turning routine behavior into an economy of XP, gold, cosmetics, and quests. Ori should borrow from all four models, but do so selectively: ship **streaks, XP ledger, levels, daily/weekly quests, achievements, cosmetic rewards, and a light social layer first**; delay punitive “hearts” or a hard energy gate until the review loop and recovery UX are excellent. [3]

If I reduce the whole report to six immediate priorities, they are these: **fix pinyin rendering and public loading shells; make the first session feel complete in under five minutes; add streaks and a transparent XP economy; ship daily and weekly quests that reward learning-forward actions rather than XP farming; add achievement badges and cosmetic rewards for delight; and build one high-repeatability mini-game, preferably a timed word-bank gauntlet tied to review words.** That package is small enough for a single full-stack developer to ship over roughly ten weeks, but it is large enough to materially improve activation, retention, and shareability. [4]

Scope and Current-State Audit

This audit is based on publicly accessible routes and public crawler snapshots, not on authenticated internal state. The snapshot evidence shows: a public landing page with demo entry points; a public path page; a public vocabulary page; a public review shell; a public profile shell; a public demo lesson route; and a dashboard shell with top navigation. Authenticated user state, database schema, real production analytics, and back-office tooling are **unknown** from public evidence. [5]

The current public product tells a coherent story in several places. The hero copy is direct: "Learn Mandarin through fast exercises." It explicitly promises short, rapid-fire drills for **pinyin, hanzi, and listening**, and it reassures the visitor that no account is needed to try the first lesson. The path page also does good work: "Starter" clearly frames beginner content such as "Meet Pinyin," "Hello!," "Goodbye," and "Greeting Review," while the overall structure hints at an HSK-aligned long arc. That is important because Duolingo's own public writing consistently stresses two things learners respond to: a **guided path** and **bite-sized, scaffolded progression**. Ori already has both at a structural level. [6]

The main public UX problems are also clear. First, there is a **credibility problem in transitional states**: `/auth` exposes only "Loading...," and `/review` exposes only "Loading review...," which gives a visitor no explanation of whether the system is waiting on auth, network, hydration, or empty review data. WCAG's guidance on status messages and error identification is relevant here: users should be told the result of an action, the waiting state of an application, or the existence of errors in text, not left with ambiguous blank states. [7]

Second, the vocabulary page has a **rendering coherence problem**. The page title and search affordance are good, but the pinyin display is visibly broken: the crawler snapshot shows many entries rendered with inserted spacing between letters, even where the intended form is a compact syllable or word. Because Unicode characters with diacritics can exist in decomposed and composed forms, and because grapheme clusters are not the same as JavaScript code units, any code that splits or animates pinyin one code point at a time risks breaking display or interaction. That makes this look less like content corruption and more like a rendering pipeline issue. [8]

Third, the public product is still thin on **felt progress**. The lesson page shows a progress counter (`1/17`) and a clear CTA ("Check"), but the public shell does not show a strong daily loop above the lesson itself. Duolingo's public feature set layers streaks, XP, quests, leagues, badges, friend streaks, reactions, and shop rewards around the core exercise loop; Memrise now highlights a progress dashboard that tracks what you learned and reviewed in the last week or month; Anki gives users due counts, review states, retention tables, and "most lapsed" prioritization. Ori's current public UX does not yet surface enough of that meta-layer. [9]

Assumptions used in the recommendations below:

- The deployed app is likely a JavaScript/TypeScript web app with client-side hydration, but the exact framework is unverified from public evidence.
- Authentication exists, but the public crawler snapshot cannot validate post-auth product depth.
- The curriculum model appears lesson-based and path-based, but the exact production schema is unknown.

- Recommendations therefore focus on **implementable interaction patterns and schema additions**, not on reverse-engineering the current database. [10]

Gamification Recommendations

The right reference point for Ori is **not** “copy Duolingo’s full surface area.” It is “select the smallest set of mechanics that increase day-one fun without harming learning.” Duolingo’s public material shows the value of streaks, XP, quests, league competition, timed challenges, gems, and friend-based accountability, but its own newer updates also reveal an important product lesson: it shifted monthly challenges to be **quest-based rather than XP-based** to reduce gaming, and it replaced Hearts with a new Energy system because mistakes-as-punishment was discouraging beginners. Memrise’s public material complements that with a softer stance on consistency dashboards “without pressure or guilt,” while Anki shows why recurrence, due-load control, and leech detection matter if the goal is retention rather than mere app activity. [11]

Mechanics comparison table

Mechanic	Recommended implementation	UX copy examples	Data model changes
Streaks	Count a day when the learner completes **one lesson or one due review set**. Add a 6-hour local grace window and one earnable “Seal Freeze” inventory item per week.	“Streak saved. See you tomorrow.” • “You kept your flame alive with one quick review.”	`user_stats.current_streak`, `best_streak`, `last_activity_date`, `grace_freezes`; `streak_events` ledge
XP economy	Use append-only XP ledger. Award XP for lesson completion, due review, perfect lesson, timed challenge, quest completion. Keep values simple and visible.	“+12 XP Lesson complete” • “+4 XP Perfect audio round”	`xp_ledger(id,user_id,source,amount,meta_json,created_a` derive totals from ledger

Mechanic	Recommended implementation	UX copy examples	Data model changes
Levels	Convert total XP into learner levels with titles themed around Ori's world. Levels should unlock cosmetics and profile flair, not gate core learning.	"Level 6 reached: Tea House Traveler"	`user_level.current_level`, or compute from XP; `level_rewar
Badges	Add milestone badges for streak, review consistency, perfect lessons, listening wins, first HSK checkpoint, and "night owl/early bird."	"Badge earned: Tone Tamer" • "Badge upgraded: Sharpshooter II"	`achievements`, `user_achievements`, `achievement_progress`
Daily quests	Three ordered quests daily: easy, medium, hard. Reward gems/seals, not just XP. Prefer learning-forward actions such as "finish next lesson," "review 10 due words," "score 80%+ in 3 lessons."	"Quest 1: Do one lesson" • "Quest 3: Clear 10 due words with 80% accuracy"	`quest_templates`, `user_daily_quests`, `quest_claims`

Mechanic	Recommended implementation	UX copy examples	Data model changes
Weekly quests	One weekly quest focused on consistency and course progress, not raw grinding. Example: finish 5 days, complete 3 path steps, review 40 due items.	“Weekly quest: Keep Ori moving for 5 days”	`user_weekly_quests`, `weekly_cycles`
Social features	Start lightweight: friend add, nudge, clap/reaction on milestones, optional friend streak. Do not start with in-app chat.	“Nudge sent: go keep our streak alive” • “Clap for Alex’s 7-day streak”	`friendships`, `social_feed_events`, `nudge_events`, `shared_streaks`
Leaderboards	Ship **opt-in** weekly cohorts of 20–30 and rank by a capped weekly “Learning Score,” not pure XP. Blend XP, due-review completion, and path progress.	“You’re #4 this week” • “Do one more lesson to hold your spot”	`leaderboard_weeks`, `leaderboard_entries`, `leaderboard_cohorts`, `opt_out` flag

Mechanic	Recommended implementation	UX copy examples	Data model changes
Timed challenges	Create one repeatable challenge first: timed word-bank gauntlet. Use combo multipliers and review-word bias.	“30 seconds. Build as many correct sentences as you can.”	`challenge_runs`, `challenge_run_items`, `best_scores`
Cosmetic rewards	Let learners buy or unlock avatars, path seal skins, profile frames, and lesson themes. Cosmetics are pure delight and should not affect learning fairness.	“New path theme unlocked: Red Lantern”	`cosmetics`, `user_cosmetics`, `equipped_cosmetics`, `shop_inventory`
Meta-progression	Tie progress to “seals,” “scrolls,” or “trails.” Reward full-unit mastery with visible seal upgrades and checkpoint ceremonies.	“Starter seal awakened” • “HSK 1 checkpoint cleared”	`user_unit_mastery`, `seal_state`, `checkpoint_completions`
Energy or hearts	**Do not ship punitive hearts in the next phase.** If you need pacing later, use “Focus Energy” as an optional bonus meter for challenge modes only.	“Focus Energy full: timed modes boosted”	Optional later: `energy_current`, `energy_max`, `energy_refills_at`

Mechanic	Recommended implementation	UX copy examples	Data model changes
Randomized rewards	Add low-stakes reward chests after daily quest completion or streak milestones. Rewards can include cosmetics shards, extra freezes, profile stickers, or seal particles.	“You opened a Moon Chest”	`reward_tables`, `reward_drops`, `inventory_items`
Season pass	Add only after the core loop is stable. A free seasonal track should reward consistent play over 4–6 weeks through quests, not purchase pressure.	“Season 1: Lantern Month”	`seasons`, `season_tracks`, `user_season_progress`, `season_rewards`

In practical terms, the **highest-return first wave** is: streaks, XP ledger, levels, daily quests, weekly quests, badges, and one timed challenge. That mix is closest to Duolingo’s public motivation stack, but Ori should copy Duolingo’s **newer restraint**, not just its earlier maximalism: use quests to push **next lesson**, **due review**, and **accuracy goals**, because Duolingo itself says it moved monthly challenges away from XP farming and carefully orders daily quests from easiest to hardest. [12]

The biggest place to be opinionated is the **energy/hearts question**. Duolingo’s own 2025 product explanation says the older Hearts system was frustrating and that beginners were **2x more likely** to run out mid-lesson; that is a strong public signal that punishing mistakes too early can damage activation. Ori should therefore postpone any hard fail-state economy until its review, retry, and recovery experience is clearly delightful. If you want tension, put it inside optional timed modes rather than the core lesson path. [13]

Coherence and UX Recommendations

Ori’s public experience is close to coherent, but not yet fully **trustworthy**. Good product coherence comes from making every layer agree: the promise on the landing page, the first lesson, the path, the review system, the vocabulary presentation, and all loading/error states. Duolingo’s public writing emphasizes a guided path, scaffolded progression, and short, clear, user-friendly writing; W3C guidance requires readable contrast, understandable status messages, clear error text, sensible focus behavior, and sufficiently large or spaced targets. Ori should aim for that standard before it adds more ornament. [14]

Coherence and UX change table

Suggestion	Exact UI and UX change	Code-level hints	Priority
Demo-first onboarding	Land every new visitor in a 3-step funnel: hero → 90-second demo lesson → "Save progress" gate. Do **not** push auth before the user feels one full win.	Route `/` CTA to `/lesson/lesson-sa-1?mode=demo`; persist demo progress in local storage; prompt auth only after lesson complete.	Highest
First-session pacing	Make lesson 1 finish in under 3 minutes and guarantee at least one early perfect answer. Front-load a listening or recognition win before typed production.	Add `lesson_intro_weight` and `first_session_only` flags to exercise ordering; cap first session to 8–10 items.	Highest
Feedback language	Replace generic correctness feedback with warm, precise coaching. Example: "Correct. 你 means 'you.'" or "Almost — 是 is the missing verb here."	Create `feedback_templates` by exercise type; support `correct_explanation`, `wrong_explanation`, `next_hint`.	High
Navigation coherence	Rename tabs to a cleaner loop: **Learn, Path, Review, Words, Me** or **Learn, Path, Practice, Words, Me**. "Review" is good, but "Learn" and "Path" may feel duplicative.	Standardize nav labels in a single enum; centralize icons and route map.	High
Loading states	Replace bare "Loading..." shells with skeletons plus plain-English explanation: "Loading your lesson..." "Checking what words are due..." "Sign in to sync your progress."	Use suspense skeletons and typed async state objects; render `loading`, `empty`, `auth_required`, `network_error` separately. Add `role=status`.	Highest

Suggestion	Exact UI and UX change	Code-level hints	Priority
Empty review state	If there is nothing due, say so proudly: “Nothing due right now. Do a new lesson to grow tomorrow’s review.”	Review page should branch on `due_count === 0`; surface CTA back to path.	High
Vocabulary coherence	Make the vocabulary page feel like a “word garden,” not a raw dump. Group by recent, weak, favorite, HSK level, lesson origin.	Add `word_tags`, `first_seen_lesson_id`, `strength_bucket`, `is_favorite`; support faceted filters.	Medium
Visual hierarchy	One primary action per screen. In lessons, only the exercise, progress, and primary CTA should dominate. Secondary hints, pinyin toggles, and audio replay should be visibly secondary.	Design token pass for typography scale, spacing, button variants; audit z-index and emphasis hierarchy.	High
Color and branding consistency	Pick one signature accent family plus one success and one danger family. Avoid multi-hue randomness across path seals, buttons, and cards unless it follows a formal token system.	Add semantic tokens: `--brand`, `--brand-2`, `--success`, `--danger`, `--warning`, `--surface`, `--surface-muted`.	Medium
Mobile-first touch design	Make every tappable choice at least comfortably thumb-sized and spaced. Word-bank tiles should not require precision dragging.	Prefer tap-to-select over drag-only; keep target size and spacing at or above WCAG minimum guidance; add single-pointer alternatives.	Highest
Accessibility pass	Ensure 4.5:1 text contrast where required, keyboard-visible focus rings, screen-reader status updates, persistent error text, and form labels.	Use semantic headings, `aria-live`/`role=status`, `aria-invalid`, visible focus tokens, logical tab order, non-color error cues.	Highest

Suggestion	Exact UI and UX change	Code-level hints	Priority
Auth flow clarity	On auth-required boundaries, explain why sign-in helps: “Save streaks, XP, and your word history.” Avoid abrupt redirects.	Add `auth_intent` context to modal and route guard; preserve return URL.	High
Pinyin rendering fix	Stop rendering pinyin character-by-character unless segmentation is grapheme-safe. Normalize to NFC before display, store a syllable array for advanced highlighting, and never build spacing with `split('')`.	`display = pinyin.normalize('NFC')`; if segmentation is needed, use `Intl.Segmenter(..., {granularity:'grapheme'})`; store `pinyin_display`, `pinyin_numeric`, `pinyin_syllables text[]`.	Highest

The pinyin bug deserves a concrete fix plan because it is both user-visible and brand-damaging. The web snapshot strongly suggests that the current render path is splitting or spacing pinyin at the wrong unit, producing `y ī` rather than `yī` and `Z h ō n g g u ó` rather than `Zhōngguó`. In JavaScript, `String.prototype.normalize()` gives you consistent Unicode normalization, and `Intl.Segmenter` gives you grapheme-aware segmentation rather than naive character splitting. Unicode normalization guidance also makes clear that combining marks may be reordered or decomposed, so the safest approach is: **store and display pinyin in NFC**, never derive visual syllables by `split('')`, and if you need per-syllable highlighting, store syllables as explicit data instead of inferring them at render time. [15]

A minimal display utility can look like this:

```
export function normalizePinyinForDisplay(input: string): string {
  return input.normalize("NFC").trim();
}

export function segmentGraphemes(input: string): string[] {
  const normalized = normalizePinyinForDisplay(input);
  const segmenter = new Intl.Segmenter("zh", { granularity: "grapheme" });
  return Array.from(segmenter.segment(normalized), s => s.segment);
}
```

For word-bank tiles or animated highlighting, prefer **word or syllable objects** over derived strings:

```
type PinyinWord = {
  display: string;         // "Zhōngguó"
  numeric?: string;        // "Zhong1guo2"
```

```
  syllables?: string[];   // ["Zhōng", "guó"]
};
```

That approach lines up with the Unicode and MDN guidance above and will eliminate a whole class of fragile front-end fixes. [16]

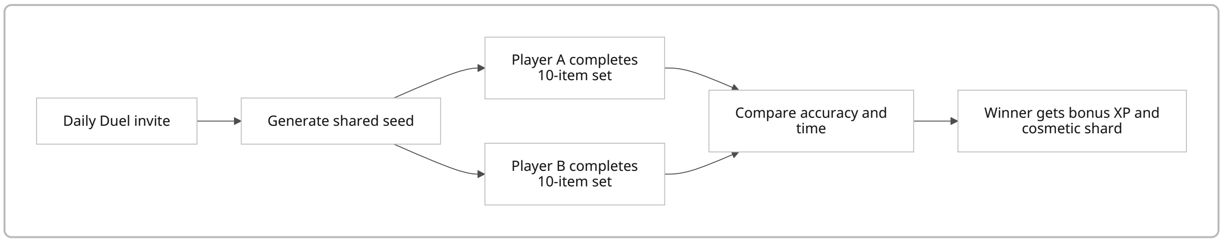
A sample “daily hub” UI that would make Ori feel much more complete on mobile:

```
┌────────────────────────────────────┐
│ Ori                                │
│ 🔥 7-day streak     ✨ 1,240 XP     │
│ Today: 2/3 quests completed        │
├────────────────────────────────────┤
│ Continue where you left off        │
│ Lesson: Meet Pinyin                │
│ [ Start 3-min lesson ]             │
├────────────────────────────────────┤
│ Due now                            │
│ 12 words need review               │
│ [ Review now ]                     │
├────────────────────────────────────┤
│ Today’s quests                     │
│ ✓ Do 1 lesson                       │
│ ✓ Review 10 words                   │
│ □ Score 80%+ in 3 lessons          │
├────────────────────────────────────┤
│ Weekly progress                    │
│ 3 / 5 active days                  │
│ [ View path ]                      │
└────────────────────────────────────┘
```

Non-course Features and Mini-games

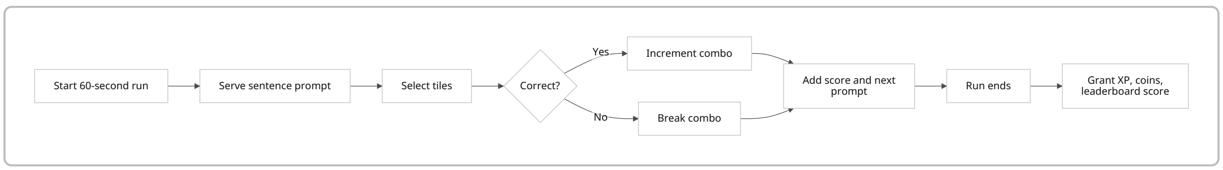
The best non-course features for Ori are the ones that **repackage review** rather than distracting from it. Duolingo’s public product includes timed challenges, side quests, friends quests, friend streaks, and achievement systems; Memrise publicly leans into activity dashboards and fixed-term challenges; Anki shows the value of filtered study modes such as “most lapses” and due-priority review; Habitica demonstrates how cosmetics, quests, and gold-like currencies can sustain engagement across repeated routines. Ori should turn its best asset—short Mandarin exercise loops—into reusable game modes. [17]

Daily Duel
An asynchronous head-to-head challenge where two learners get the same 10-item review set and compete on score plus speed. This is Ori’s safest social mode because it reuses existing exercises and avoids real-time multiplayer complexity.



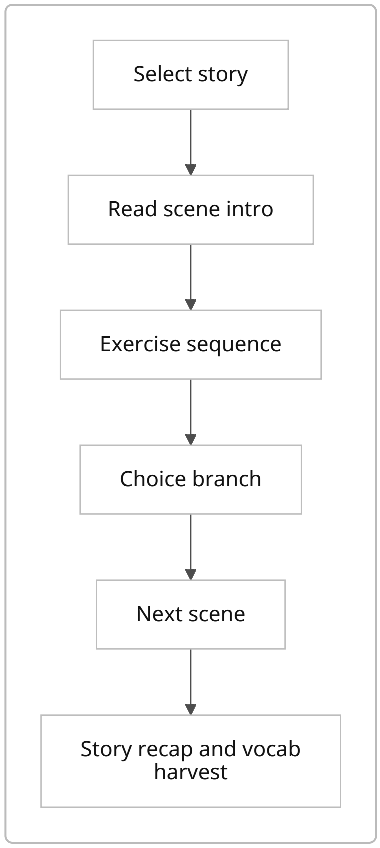
Schema additions: `duel_matches`, `duel_rounds`, `duel_participants`, `duel_seed`, `duel_scores`.
XP and economy: base XP for participation, bonus XP for win, 1 cosmetic shard for first duel win of the day.

Timed Word Bank Gauntlet
A fast, repeatable mode built around word-bank sentence assembly with combos and multipliers. This should be Ori's first true mini-game because it is high-repeatability, language-real, and easy to explain.



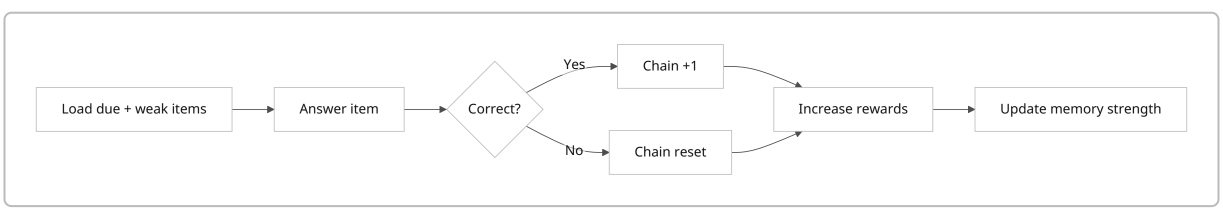
Schema additions: `challenge_runs`, `challenge_scores_daily`, `combo_stats`, `best_run_by_user`.
XP and economy: XP based on score bands; top daily personal score grants coins or seal dust.

Story Mode
Short micro-stories that chain 4–6 exercises into a scene, such as ordering tea, introducing yourself, or taking a taxi. Duolingo's public path design and story integration make this a strong medium-term fit. [18]



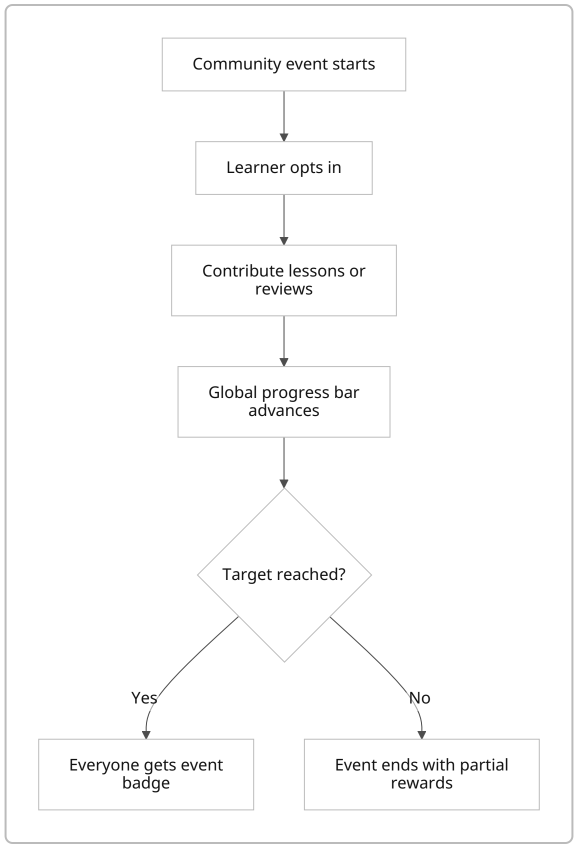
Schema additions: `stories`, `story_nodes`, `story_branches`, `story_rewards`, `story_attempts`.
XP and economy: higher XP than a standard lesson; unlock a story badge and collectible postcard art after completion.

Recall Chain
A pure review mode for due and weak words where every correct answer extends a chain and every mistake drops the multiplier. This is the closest Ori can get to making Anki-style due review feel exciting rather than dutiful. Anki's own manual explicitly supports due sorting, "most lapses," and leech handling; Ori can gamify that same logic. [19]



Schema additions: `recall_chain_runs`, `memory_lapse_count`, `leech_flags`, `review_priority_snapshot`.
XP and economy: XP multiplier tied to chain length; extra coins for clearing all flagged weak items.

Community Challenge
A fixed-time event, usually weekend-based, with a single shared goal such as "collectively finish 50,000 review words." Memrise's public 30-day challenge and Duolingo's quest/event framing both support the idea that limited-time, visible participation increases engagement. [20]

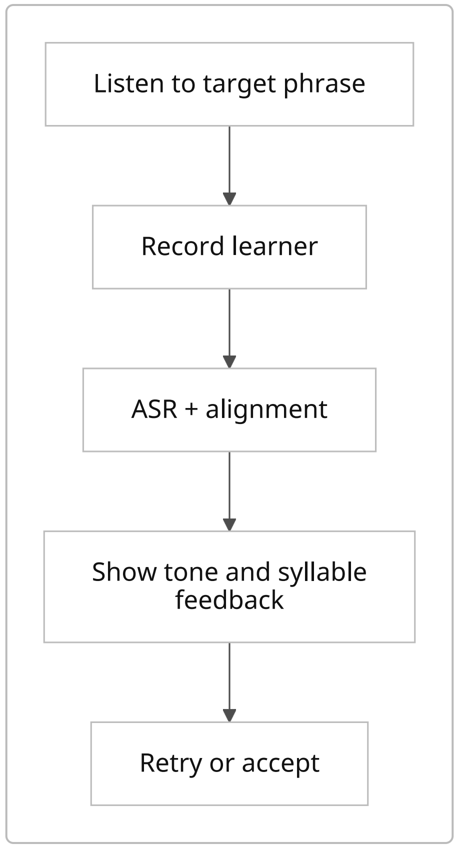


Schema additions: `community_events`, `community_event_entries`, `community_progress_aggregate`, `event_rewards`.
XP and economy: normal XP plus event tokens; badge if personal minimum contribution met.

Custom Decks
Let learners create auto-decks such as "Words I missed this week," "Food," "HSK 1 only," or "My favorites." This is the most obviously Anki-like feature and would deepen Ori's power-user appeal without changing the beginner path. [21]

Schema additions: `custom_decks`, `custom_deck_rules`, `custom_deck_items`, `custom_deck_attempts`.
XP and economy: lower XP than primary path lessons to discourage farming, but great for badge progress.

Pronunciation Lab
Not for the immediate next sprint, but worth designing now. Memrise's public material still argues that speaking confidence matters and AI conversation/speaking tools are part of its stack, while the user's own preference is that speaking should be better than Duolingo's weak implementation. Ori should therefore treat speaking as a later, separate lab mode, not as a core lesson mechanic yet. [22]



Schema additions: `pronunciation_attempts`, `asr_scores`, `tone_error_types`, `audio_assets`.
XP and economy: modest XP, plus specific speaking badges; no leaderboard until scoring is trustworthy.

Roadmap, Analytics, and MVP+ Recommendation

A realistic Cursor-friendly roadmap for a single full-stack developer is **10 weeks at roughly 30–38 focused dev-hours per week**. That is enough time to land meaningful retention features without overreaching into multiplayer chat, full season systems, or production ASR. The roadmap below assumes the current deployment already has the lesson engine, course path, vocabulary page, review shell, and auth surface in place. [23]

Ten-week implementation roadmap

Week	Milestone	Deliverables	Acceptance criteria	Rough hours
Week one	Activation and trust pass	Fix pinyin rendering; replace all public bare loading states; add auth-required and empty-review states; add error logger and product analytics scaffold	Vocabulary pinyin renders as compact syllables; `/auth` and `/review` no longer show bare loading text; session analytics flow end-to-end	34
Week two	Demo-first onboarding	Rewrite landing CTA flow into demo → save-progress funnel; shorten first session; add lesson-complete modal with save prompt	New users can complete a meaningful demo in under 3 minutes and are offered sign-up only after completion	32
Week three	Streak and XP foundation	XP ledger, streak counter, best streak, weekly XP summary, post-lesson reward toasts	Every lesson and review writes XP ledger entries; streak updates correctly across timezone boundaries and grace window	36
Week four	Levels and badges	Level titles, milestone badges, profile badge shelf, first 12 achievements	Users receive badge toasts and progress persists correctly; profile shows earned and in-progress badges	34
Week five	Daily and weekly quests	3 daily quests, 1 weekly quest, quest rewards, claim flow, quest progress surfaces on dashboard	Quests roll over on schedule; completion rewards are claimable; no quest can be completed by idle visits only	38
Week six	Review depth	Recall Chain mode, weak-word prioritization, lapse counter, leech tagging, improved due sorting	Review surface can prioritize due, weak, and “most lapsed” items; user can finish a Recall Chain run and see memory updates	36
Week seven	Mini-game launch	Timed Word Bank Gauntlet, combo system, personal bests, daily challenge score	Gauntlet is playable on mobile and desktop; score, combo, and XP payout are deterministic and logged	38
Week eight	Social groundwork	Friend model, add/remove friend, milestone clap, nudge, optional shared streak beta	A learner can add a friend, send a nudge, and react to a milestone; no freeform chat exists	34

Week	Milestone	Deliverables	Acceptance criteria	Rough hours
Week nine	Community and cosmetics	Cosmetic shop, profile frames/ path themes, small randomized reward chest, weekend community challenge beta	Reward drops and purchases persist; event participation and contribution tracking work correctly	36
Week ten	Polish and validation	A/B tests, dashboard redesign, accessibility sweep, instrumentation QA, launch checklist	Core paths meet contrast/ focus/status requirements; experiments ship behind flags; dashboards reflect live data	32

Analytics events to instrument

Event name	When it fires	Key properties
`landing_viewed`	Landing page render	`utm_source`, `device_type`, `signed_in`, `ab_variant`
`demo_started`	First demo lesson begins	`lesson_id`, `source_cta`, `is_demo`, `device_type`
`demo_completed`	Demo lesson completion	`lesson_id`, `duration_sec`, `score_pct`, `mistakes`, `signup_prompt_shown`
`signup_prompt_viewed`	Post-demo auth gate appears	`context`, `streak_preview`, `xp_preview`
`auth_completed`	User account created or signed in	`method`, `return_to`, `from_demo`
`lesson_started`	Standard lesson begins	`lesson_id`, `unit_id`, `course_id`, `exercise_count`, `source`
`exercise_answered`	Each submission	`exercise_type`, `is_correct`, `attempt_index`, `latency_ms`, `used_hint`, `had_audio`
`lesson_completed`	Standard lesson finished	`lesson_id`, `score_pct`, `duration_sec`, `xp_awarded`, `perfect`
`review_started`	Review mode begins	`review_mode`, `due_count`, `weak_count`
`review_completed`	Review session ends	`items_seen`, `accuracy_pct`, `xp_awarded`, `memory_items_updated`

Event name	When it fires	Key properties
`streak_extended`	Day counts toward streak	`current_streak`, `best_streak`, `used_freeze`
`quest_progressed`	Any quest progress increment	`quest_id`, `quest_type`, `progress`, `target`
`quest_claimed`	Reward claimed	`quest_id`, `reward_type`, `reward_value`
`challenge_started`	Mini-game starts	`challenge_type`, `seed`, `entry_cost`
`challenge_completed`	Mini-game ends	`challenge_type`, `score`, `best_score`, `xp_awarded`
`badge_earned`	Achievement unlocked	`badge_id`, `badge_tier`, `surface`
`friend_nudged`	Social nudge sent	`target_user_id_hash`, `context`, `shared_streak`
`leaderboard_viewed`	Leaderboard opened	`cohort_size`, `rank`, `opt_in`
`loading_error_seen`	User sees a retriable error	`route`, `error_type`, `network_state`

A/B tests worth running

Test	Variant A	Variant B	Success metric	Guardrail metric
Demo CTA	“Try demo”	“Start your first 3-minute lesson”	Demo start rate	Bounce rate
Save gate timing	Ask for sign-up before results	Ask after lesson-complete rewards	Auth conversion	Demo completion rate
Daily quest structure	XP-focused quests	Progress-focused quests	D7 retention	Review completion rate
Streak requirement	One lesson/day	One lesson **or** 10 due reviews/day	D14 retention	Review quality and average session length
Reward style	XP-only completion toast	XP + cosmetic shard/ chest moments	Sessions/ week	Lesson completion time
Mini-game surfacing	Dedicated tab	Embedded in dashboard after one lesson	Challenge adoption	Path lesson starts

Test	Variant A	Variant B	Success metric	Guardrail metric
Leaderboard scoring	Pure weekly XP	Blended learning score	Weekly active days	Review completion share

Final recommended six-item MVP+ roadmap

The six-item plan I would actually recommend committing to first is this:

1. **Fix trust breakers first**: pinyin rendering, public loading shells, and explicit empty/error/auth states.
2. **Make demo-to-signup excellent**: a short first win, a satisfying result screen, and a clear reason to create an account.
3. **Ship the habit core**: streaks, XP ledger, and levels.
4. **Ship learning-aligned quests**: three daily quests and one weekly quest centered on path progress and due review, not grinding.
5. **Ship delight**: achievements plus cosmetic rewards, because they are highly visible and low-risk.
6. **Ship one sticky repeatable mini-game**: the Timed Word Bank Gauntlet, tied directly to weak and due words.

That six-item package is the best balance of **effort, retention upside, and product coherence** for Ori's current stage. It strengthens what the live MVP already demonstrates publicly—structured content, a real lesson engine, and a clear Mandarin focus—while addressing the main things the current public experience still lacks: visible progress, stronger repeat motivation, cleaner polish, and a higher-quality first impression. [24]

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[1] [6] [10] [14] [23] [24] Ori — Learn Mandarin
https://ori-sepia-nine.vercel.app/?utm_source=chatgpt.com

[2] [7] Ori — Learn Mandarin
https://ori-sepia-nine.vercel.app/auth?utm_source=chatgpt.com

[3] [11] The habit-building research behind your Duolingo streak
https://blog.duolingo.com/how-duolingo-streak-builds-habit/?utm_source=chatgpt.com

[4] [8] [15] Ori — Learn Mandarin
https://ori-sepia-nine.vercel.app/vocabulary?utm_source=chatgpt.com

[5] Ori — Learn Mandarin
https://ori-sepia-nine.vercel.app/dashboard?utm_source=chatgpt.com

[9] Ori — Learn Mandarin
https://ori-sepia-nine.vercel.app/lesson/lesson-sa-1?utm_source=chatgpt.com

[12] Understanding Duolingo's Time Spent Learning Well Metric
https://blog.duolingo.com/time-spent-learning-well/?utm_source=chatgpt.com

[13] Why Duolingo switched to Energy and how it helps you learn
https://blog.duolingo.com/duolingo-energy/?utm_source=chatgpt.com

16 String.prototype.normalize() - JavaScript - MDN Web Docs
https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/String/normalize?utm_source=chatgpt.com

17 How to Use Duolingo for Language Learning
https://blog.duolingo.com/duolingo-101-how-to-learn-a-language-on-duolingo/?utm_source=chatgpt.com

18 Introducing the new Duolingo learning path
https://blog.duolingo.com/new-duolingo-home-screen-design/?utm_source=chatgpt.com

19 Filtered Decks - Anki Manual
https://docs.ankiweb.net/filtered-decks.html?utm_source=chatgpt.com

20 The Memrise 30-Day Challenge
https://www.memrise.com/blog/30-day-challenge-2025?utm_source=chatgpt.com

21 Getting Started - Anki Manual - AnkiWeb
https://docs.ankiweb.net/getting-started.html?utm_source=chatgpt.com

22 About us
https://www.memrise.com/about?utm_source=chatgpt.com